

A Residual Algebra for Catalytic Composition: Derived Floors, the Telescoping Obstruction, and Closure as a Termination Criterion

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Abstract

We develop an algebra for events that reduce the uncertainty of a bounded observer about a target, and for the composition of such events into cascades. The algebra is built on one quantity, the *residual*: the irreducible remainder of an identification that cannot be completed. We prove that the residual is strictly positive whenever identification proceeds by comparison against a whole that no finite stage exhausts, so that the positive floor central to the algebra is *derived* rather than posited. On this floor we define the *catalytic power* κ of an event as the fraction of the remaining distance to the floor that the event closes, and prove that catalytic powers compose by the multiplicative law $\kappa(\gamma_1 \diamond \gamma_2) = 1 - (1 - \kappa_1)(1 - \kappa_2)$, that composition is associative and commutative with identity, that the floor is never attained in finitely many steps, and that convergence to the floor under an infinite cascade holds if and only if $\sum_i \kappa_i = \infty$. We then isolate an obstruction that invalidates the obvious empirical test of the composition law: when the powers of the constituent events are estimated from the same state sequence as the cascade outcome, the multiplicative prediction and the measurement are the *same algebraic expression*, and any correlation between them is an identity rather than evidence. We prove this (*Telescoping Theorem*), give the exact condition under which it holds, and prove that replacing instance-specific powers by type-averaged powers breaks the identity, so that the resulting test has a genuine null. We characterise the residual variance decomposition that determines the power of that test, and show the test is informative precisely when between-type variance of κ exceeds within-type variance. Finally we define *closure*—a termination criterion under which a cascade is complete when no further available event can carry the observer into a state-equivalence class not already reached—and prove that closure is strictly stronger than any fixed threshold on attained uncertainty, that every cascade over a finite event registry reaches closure in finitely many steps, and that closure resolves into exactly two outcomes: convergence on a single class, or a contested outcome whose honest report is a declared plurality of classes rather than a selected representative. All results are proved from stated definitions, and all are checked numerically against constructed instances; the accompanying computations report both the confirmations and the one regime in which the composition test is shown to have no power.

Keywords: residual algebra; catalytic composition; multiplicative law; derived floor; telescoping identity; type-averaged estimation; closure; termination criteria; bounded observers.

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1 Introduction

1.1 The question

Consider an observer that reduces its uncertainty about a target by a sequence of events: an experiment that narrows a hypothesis space, a measurement that discriminates among candidate states, a physiological transition that increases the coherence of a system, an inferential step that rules out alternatives. Each event closes part of a gap. The question this paper answers is: *how do such events compose?* If one event closes a known fraction of the gap and a second event closes a known fraction, what fraction does the pair close, and what law governs the sequence of many?

The question has an obvious candidate answer and a less obvious correct one. The obvious answer is that the reductions add. The correct answer, for any process in which each event acts on the gap that remains after its predecessors, is that the *residuals* multiply: what survives two events is what survives the first, further reduced by the second, so the surviving fractions compose by product and the closed fractions compose by $1 - (1 - \kappa_1)(1 - \kappa_2)$. This is not a modelling choice. It follows from the definition of “fraction of the remaining gap” together with the requirement that the gap be measured from a fixed floor, and we prove it in Section 5.

The substance of the paper lies in three places beyond this law. First, the floor from which the gap is measured cannot be posited without cost: if it is set to zero the algebra degenerates and the powers are undefined at convergence; if it is set to an arbitrary positive constant the algebra is parameter-dependent and its predictions unfalsifiable. We show (Section 3) that a strictly positive floor is *forced* by the structure of identification itself, provided identification proceeds by comparison against a whole that no finite stage exhausts.

Second, the composition law admits an apparently decisive empirical test that is in fact vacuous. If the powers of individual events are estimated from the same sequence of states from which the cascade outcome is measured, the multiplicative prediction reduces, by cancellation, to the measurement itself; the two agree exactly and the agreement carries no information. We state

and prove this obstruction (Section 6) because it is easy to encounter and easy to mistake for a striking confirmation. We then show what must be changed for the test to have content.

Third, a cascade must know when to stop. The usual criterion—continue until attained uncertainty falls below a threshold—is provably inadequate, because a single internally consistent line of evidence can satisfy any threshold while an unexamined alternative line would have reached an incompatible conclusion. We define an alternative criterion (Section 8), prove it strictly stronger, and prove that it terminates with exactly two possible verdicts, one of which is an honest declaration that no single answer is supported.

1.2 What is assumed

We assume one structural fact, stated precisely in Axiom 1: that identification of a part proceeds by comparison against the rest, and that the rest is not exhausted at any finite stage. Everything else is derived. In particular the positive floor, which in many treatments is an axiom, is here a theorem (Theorem 3.4). We do not assume any probability measure, any prior, any noise model, or any parametric family. The algebra is measure-free: its single quantity is a separation cost on a finite weighted graph, and its single law is a product.

1.3 Relation to existing work

The multiplicative structure we derive is formally the survival-function composition familiar from reliability theory [2], and the geometric convergence it produces is the standard behaviour of a contraction [1]. What distinguishes the present treatment is that the contraction factor is not fitted: it is a ratio of separation costs on a graph whose floor is derived. The residual quantity we use is a minimum cut, and we rely on the max-flow–min-cut correspondence and Menger’s theorem for its structural properties [6, 8]. The information-theoretic reading of untracked residuals in Section 9 uses Shannon entropy in its original sense [4, 10]. The termination criterion of Section 8 is related in spirit to the stopping rules of optimal experimental design [3, 7], but differs essentially: those rules stop when expected information gain falls below a threshold, whereas closure stops when the reachable set of *outcomes* ceases to grow, a condition that is not a function of any gain magnitude. The distinction between a threshold criterion and an exhaustion criterion is made precise in Theorem 8.3. Our insistence on reporting a contested outcome rather than a selected representative parallels the literature on conflicting evidence in belief functions [5, 9], though we reach it by a different route and without a combination rule.

1.4 Organisation

Section 2 fixes the setting. Section 3 derives the floor. Section 4 defines catalytic power and establishes its elementary properties. Section 5 proves the composition law and its consequences, including the convergence criterion. Section 6 proves the telescoping obstruction and the type-averaged correction. Section 7 characterises when the corrected test has statistical power. Section 8 develops closure. Section 9 gives the three readings of a residual. Section 10 reports numerical verification. Section 11 states scope and limitations.

2 The setting

We work with finite weighted graphs throughout. This is not a restriction on the phenomena modelled but a commitment about the observer: an observer with finite descriptive resources represents its situation by finitely many distinctions, and each distinction costs something to draw.

Definition 2.1 (Finite weighted graph; cut). A *finite weighted graph* is a triple $G = (V, E, w)$ with V finite, $E \subseteq \binom{V}{2}$, and $w: E \rightarrow (0, \infty)$. For $U \subseteq V$ the *cut* of U is $\partial(U) = \{\{u, v\} \in E : u \in U, v \notin U\}$, and its weight is $w(\partial(U)) = \sum_{e \in \partial(U)} w(e)$.

Definition 2.2 (Contact graph; medium). A *contact graph* is a connected finite weighted graph G together with a distinguished vertex $\mathfrak{M} \in V$, the *medium*, representing the rest of the whole against which parts are told apart. Vertices other than $\mathfrak{M}$ are *items*. An edge $\{u, v\}$ records that u and v have been separated, and $w(\{u, v\})$ is the cost of that separation.

Remark 2.3 (The medium is the rest, not a background). The medium is not an environment in which items sit; it is the totality of what an item is not, considered as a single vertex. Its role is to be the counterparty of every identification. An item is not individuated against some items and not others: it is individuated against everything else at once, and the cost of doing so is the weight of the cheapest cut separating it from $\mathfrak{M}$.

Definition 2.4 (Separation cost; resting cut). For an item v , the *separation cost* is

$$\sigma(v) = \min\{w(\partial(U)) : v \in U \subseteq V \setminus \{\mathfrak{M}\}\},$$

the minimum weight of a cut separating v from the medium. A minimising $U^*(v)$ is a *resting side*, and $\partial(U^*(v))$ the *resting cut*. The minimum is attained because V is finite.

Separation cost is the fundamental observable of the algebra. By the max-flow–min-cut theorem [6] it equals the maximum flow from v to $\mathfrak{M}$, and by Menger’s theorem [8] it counts, in the unweighted case, the maximum number of edge-disjoint paths that must be severed to disconnect v from the rest. Both readings support the same intuition: $\sigma(v)$ measures how much must be committed to hold v apart from everything it is not.

Definition 2.5 (State; uncertainty). A *state* of an observer with respect to a target is a contact graph G together with a designated item v . The observer’s *uncertainty* in that state is $S := \sigma(v)$, the separation cost of the target in the observer’s current graph. A state with a smaller separation cost is one in which the target is held apart more cheaply, i.e. is better individuated.

Remark 2.6 (Why cost decreases as knowledge increases). It may seem paradoxical that better individuation corresponds to *lower* cost. The reading is this: an observer that has already drawn many distinctions in the neighbourhood of v needs to commit little further to hold v apart, because most of the separating work is already done and the remaining cheapest cut is small. An observer that has drawn few distinctions must commit a great deal at once. The quantity $\sigma(v)$ therefore measures the *outstanding* commitment, and it is this that events reduce.

3 The floor, derived

The algebra requires a fixed lower bound β from which gaps are measured. We now show this bound need not be assumed.

Definition 3.1 (Expanding family; non-completable whole). An *expanding family* is a sequence of contact graphs $G_0 \subseteq G_1 \subseteq G_2 \subseteq \dots$, each obtained from its predecessor by adjoining items and contacts, representing the progressive elaboration of an observer’s distinctions. The whole is *non-completable* if for every finite stage k there is an item present in some later stage but absent from G_k : no finite stage exhausts what there is to be told apart.

Remark 3.2 (An order condition, not a cardinality claim). Non-completeness says only that the process of elaboration has no terminal stage. It does not assert that the whole is a completed infinite set, and no argument below requires such an assertion. This matters: the results are about the unavailability of a finishing point to a finite observer, which is an order-theoretic fact about the family, not a metaphysical one about the world.

Axiom 1 (Identification against a non-completable whole). Items are identified by comparison against the medium, and the whole is non-completable in the sense of Theorem 3.1.

Definition 3.3 (Identification residual). To *identify* an item v at stage k is to determine it by comparison against $\mathfrak{M}$ in G_k . The identification is *complete* if every item of the whole has been brought to bear on the comparison. The *identification residual* $\rho(v)$ is the content of the comparison that remains outstanding; a complete identification has $\rho(v) = 0$.

Theorem 3.4 (Floor from non-completeness). *Under Axiom 1, no identification of an item is complete at any finite stage, and consequently $\rho(v) > 0$ for every item v .*

Proof. Suppose $\rho(v) = 0$ at some finite stage k , so the identification of v is complete in G_k . By Theorem 3.3, completeness requires that every item of the whole has been brought to bear on the comparison of v against $\mathfrak{M}$; every such item is therefore present in G_k . Hence G_k contains the whole, contradicting non-completeness (Theorem 3.1), which supplies an item absent from G_k . Therefore no finite stage completes the identification and $\rho(v) > 0$. $\square$

Corollary 3.5 (Positive floor). *There is a constant $\beta > 0$ with $\sigma(v) \geq \beta$ for every item v of every stage, and $w(e) \geq \beta$ for every contact e .*

Proof. Put $\beta := \inf_v \rho(v)$ over the items of the family. By Theorem 3.4 each $\rho(v) > 0$. If $\beta = 0$ there are items with residual arbitrarily close to zero; such an identification is arbitrarily close to complete, and in the limit completes, which Theorem 3.4 forbids. Hence $\beta > 0$. The residual of v is carried by the contacts of its resting cut, so $\sigma(v) \geq \beta$; and each contact $\{u, v\}$ lies in the cut of the singleton $\{u\}$, whence $w(e) \geq \beta$. $\square$

Corollary 3.6 (No sharp identification). *No item is individuated at zero cost, and no state has $S = 0$. The attainable values of the uncertainty S lie in $[\beta, \infty)$.*

Remark 3.7 (Status of the floor). The floor is not a limitation of an instrument and not a tunable constant. It is the outstanding remainder of a comparison that cannot be finished, and its positivity is exactly the statement that the comparison cannot be finished. Two consequences are used repeatedly below: the denominator $S - \beta$ appearing in Theorem 4.1 is well defined and positive for every attainable state above the floor; and the floor is a property of the observer's situation rather than of any particular event, so it is held fixed across a cascade.

Remark 3.8 (The floor is estimable, and its positivity is falsifiable). Theorem 3.4 predicts an empirical signature. If the floor is genuine, then estimating $\sigma(v)$ as the family expands—as more of the whole is brought to bear—should exhibit convergence to a strictly positive asymptote rather than decay toward zero. This is a testable claim and is not guaranteed by construction.

We stress the contrast with an estimator that defines the floor as the minimum observed value of S in a finite sample. Such an estimator is strictly positive whenever the sample is, so it can never return a value inconsistent with a positivity claim: it has no capacity to falsify. It is moreover biased upward, a sample minimum being an upper bound on the infimum of the underlying process, and by Theorem 5.10 an upward-biased floor inflates precisely the powers of the events acting nearest the floor. The relevant distinction is between *falsification* and *ordering*: given unboundedly many draws from a process whose density approaches zero, a sample minimum eventually becomes small and may then order an unfloored process below a floored one. That ordering is not a test of positivity, and in the finite-sample regime an experimenter actually occupies, the sample minimum reports a confidently positive floor for a process that has none. Section 10 implements both estimators in a realistic and a large-sample regime and reports the contrast.

4 Catalytic power

Definition 4.1 (Event; catalytic power). An *event* γ carries an observer from a state with uncertainty $S_{\text{before}} > \beta$ to a state with uncertainty S_{after} . Its *catalytic power* is

$$\kappa(\gamma) = \frac{S_{\text{before}} - S_{\text{after}}}{S_{\text{before}} - \beta}$$

the fraction of the outstanding gap to the floor that the event closes.

Definition 4.2 (Residual factor). The *residual factor* of an event is $\bar{\kappa}(\gamma) := 1 - \kappa(\gamma)$, the fraction of the gap that survives it. Explicitly

$$\bar{\kappa}(\gamma) = \frac{S_{\text{after}} - \beta}{S_{\text{before}} - \beta}.$$

The residual factor is the natural coordinate of the algebra: the entire composition theory is the statement that residual factors multiply, and every result below is more transparent in $\bar{\kappa}$ than in κ .

Proposition 4.3 (Elementary properties). *Let γ be an event acting on a state with $S_{\text{before}} > \beta$. Then:*

- (i) $\kappa(\gamma) \leq 1$, with equality iff $S_{\text{after}} = \beta$;
- (ii) $\kappa(\gamma) > 0$ iff the event strictly reduces uncertainty, and $\kappa(\gamma) < 0$ iff it strictly increases it;
- (iii) $\kappa(\gamma) = 0$ iff the event leaves uncertainty unchanged;
- (iv) $\bar{\kappa}(\gamma) \geq 0$ for every event whose post-state is attainable, by Theorem 3.6.

Proof. All four are immediate from Theorems 4.1 and 4.2 and the positivity of the denominator $S_{\text{before}} - \beta$. For (i), $\kappa \leq 1$ is equivalent to $S_{\text{after}} \geq \beta$, which is Theorem 3.6; equality holds exactly when $S_{\text{after}} = \beta$. For (iv), $\bar{\kappa} = (S_{\text{after}} - \beta)/(S_{\text{before}} - \beta) \geq 0$ since both numerator and denominator are nonnegative, the numerator by Theorem 3.6. $\square$

Remark 4.4 (Negative powers are admissible and meaningful). Nothing forces $\kappa \geq 0$. An event that increases the outstanding commitment—one that draws a distinction which makes the target harder to hold apart, or that introduces confusable alternatives—has $\kappa < 0$ and residual factor exceeding one. The algebra accommodates such events without modification, and Section 10 exhibits data in which they occur systematically: transitions away from a dominant attractor state carry large negative power. Excluding them would amount to assuming that every event is informative, which is exactly what an experimental apparatus should not assume.

Proposition 4.5 (Scale invariance). *Catalytic power is invariant under affine reparametrisation of the uncertainty that fixes the floor: for $a > 0$ and $S \mapsto a(S - \beta) + \beta$, every $\kappa(\gamma)$ is unchanged.*

Proof. Both numerator and denominator of Theorem 4.1 scale by a : $(a(S_{\text{b}} - \beta) - a(S_{\text{a}} - \beta)) / (a(S_{\text{b}} - \beta)) = (S_{\text{b}} - S_{\text{a}}) / (S_{\text{b}} - \beta)$. $\square$

Theorem 4.5 is what makes the algebra portable across substrates. The units of S are arbitrary; only the position of a state relative to the floor matters. It also identifies precisely the one quantity that must be determined per observer before any power can be computed—the floor—and explains why an error in the floor is not a change of units but a distortion of every power in the cascade. We return to this sensitivity in Theorem 5.10.

5 Composition

Definition 5.1 (Cascade). A *cascade* is a finite sequence $\gamma_1, \dots, \gamma_n$ of events applied in order to an observer with fixed floor β , with the post-state of γ_i the pre-state of γ_{i+1} . Writing S_1 for the initial and S_{n+1} for the terminal uncertainty, the *net power* of the cascade is $\kappa_{\text{net}} = (S_1 - S_{n+1}) / (S_1 - \beta)$, i.e. the power of the cascade regarded as a single event.

Theorem 5.2 (Multiplicative composition). *For any cascade $\gamma_1, \dots, \gamma_n$ over a fixed floor,*

$$\kappa_{\text{net}} = 1 - \prod_{i=1}^n (1 - \kappa(\gamma_i))$$

equivalently, the residual factors multiply: $\bar{\kappa}_{\text{net}} = \prod_{i=1}^n \bar{\kappa}(\gamma_i)$.

Proof. Write $g_i := S_i - \beta$ for the gap before event γ_i , so $g_i > 0$ for $i \leq n$ by hypothesis and $g_{n+1} \geq 0$ by Theorem 3.6. By Theorem 4.2, $\bar{\kappa}(\gamma_i) = g_{i+1}/g_i$. Hence

$$\prod_{i=1}^n \bar{\kappa}(\gamma_i) = \prod_{i=1}^n \frac{g_{i+1}}{g_i} = \frac{g_{n+1}}{g_1} = \frac{S_{n+1} - \beta}{S_1 - \beta} = 1 - \kappa_{\text{net}},$$

the last equality by Theorem 5.1. Rearranging gives the displayed formula. $\square$

Remark 5.3 (What the law says, and what it does not). Theorem 5.2 is a consequence of the definition of catalytic power as a fraction of an outstanding gap measured from a fixed floor. It is therefore not, by itself, an empirical claim: given the definitions, it cannot fail. What *is* empirical is whether a proposed decomposition of a process into events, with powers estimated independently of the outcome being predicted, obeys it. This distinction is the entire subject of Section 6, and neglecting it is the error that section exists to forestall.

Corollary 5.4 (Algebraic structure). *Composition $\diamond$ defined by $\kappa_1 \diamond \kappa_2 := 1 - (1 - \kappa_1)(1 - \kappa_2)$ makes $(-\infty, 1]$ a commutative monoid with identity 0 and absorbing element 1. It is associative and commutative, and $\kappa \diamond 0 = \kappa$, $\kappa \diamond 1 = 1$.*

Proof. The map $\kappa \mapsto 1 - \kappa$ is a bijection $(-\infty, 1] \rightarrow [0, \infty)$ carrying $\diamond$ to ordinary multiplication, which is associative and commutative with identity 1 and absorbing element 0. Transporting back, $\diamond$ is associative and commutative with identity 0 and absorbing element 1. $\square$

Remark 5.5 (Order independence and its limits). Commutativity in Theorem 5.4 is a statement about the composition of *given* powers, not a claim that events may be reordered in practice. Reordering a physical cascade may change the powers themselves, since $\kappa(\gamma)$ is defined relative to the state on which γ acts. What the corollary says is that if the powers are what they are, the net effect does not depend on the order in which they are multiplied. The empirical content of order-independence—that the same event has the same power wherever it occurs in a cascade—is a separate hypothesis, and is exactly what the type-averaged test of Section 6 probes.

Proposition 5.6 (Diminishing returns). *For a cascade of events with powers $\kappa_i \in (0, 1)$, the net power is strictly increasing in each κ_i and satisfies $\kappa_{\text{net}} < 1$ for every finite n . The marginal contribution of γ_{n+1} to the net power is $\kappa_{n+1} \prod_{i \leq n} (1 - \kappa_i)$, which decreases geometrically in n whenever the powers are bounded below by a positive constant.*

Proof. Monotonicity is clear from $\partial \kappa_{\text{net}} / \partial \kappa_j = \prod_{i \neq j} (1 - \kappa_i) > 0$. Since each $1 - \kappa_i > 0$, the product is positive and $\kappa_{\text{net}} < 1$. For the marginal contribution, compose the first n events into $\kappa_{\text{net}}^{(n)}$; then $\kappa_{\text{net}}^{(n+1)} - \kappa_{\text{net}}^{(n)} = (1 - \kappa_{\text{net}}^{(n)}) \kappa_{n+1} = \kappa_{n+1} \prod_{i \leq n} (1 - \kappa_i)$. If $\kappa_i \geq c > 0$ then this is at most $(1 - c)^n$, which decays geometrically. $\square$

Corollary 5.7 (The floor is not attained in finite time). *No finite cascade of events with powers $\kappa_i < 1$ brings the uncertainty to the floor. The gap after n events is $g_1 \prod_{i \leq n} (1 - \kappa_i) > 0$.*

Theorem 5.8 (Convergence criterion). *Let $\gamma_1, \gamma_2, \dots$ be an infinite cascade with powers $\kappa_i \in [0, 1)$. Then the uncertainty converges to the floor, $S_n \rightarrow \beta$, if and only if*

$$\sum_{i=1}^{\infty} \kappa_i = \infty.$$

Proof. The gap after n events is $g_{n+1} = g_1 \prod_{i \leq n} (1 - \kappa_i)$, so $S_n \rightarrow \beta$ iff $\prod_{i=1}^{\infty} (1 - \kappa_i) = 0$. For $\kappa_i \in [0, 1)$ take logarithms: $\prod (1 - \kappa_i) = 0$ iff $\sum_i -\log(1 - \kappa_i) = \infty$. Since $-\log(1 - x) \geq x$ for $x \in [0, 1)$, the divergence of $\sum \kappa_i$ implies divergence of the log-sum. Conversely if $\sum \kappa_i < \infty$ then $\kappa_i \rightarrow 0$, and for $x \in [0, \frac{1}{2}]$ one has $-\log(1 - x) \leq 2x$, so the log-sum is bounded by a constant plus $2 \sum \kappa_i < \infty$, whence the product is strictly positive. Thus $\prod (1 - \kappa_i) = 0$ iff $\sum \kappa_i = \infty$. $\square$

Remark 5.9 (Interpretation of the criterion). Theorem 5.8 has the form of a Borel–Cantelli dichotomy: what matters is not the size of any individual power but whether the powers are summable. A cascade of events each closing a rapidly shrinking fraction of the gap—say $\kappa_i = 2^{-i}$ —never reaches the floor even in the limit, while a cascade of arbitrarily weak but non-summable events— $\kappa_i = 1/i$ —does. This distinguishes processes that stall from processes that converge, using only the sequence of powers.

Proposition 5.10 (Sensitivity to the floor). *Let the true floor be β and let an analysis use $\beta' = \beta + \varepsilon$ with ε small relative to the gap. Then the power computed for an event with true power κ is*

$$\kappa' = \kappa \cdot \frac{S_b - \beta}{S_b - \beta - \varepsilon} = \kappa \left(1 + \frac{\varepsilon}{S_b - \beta} \right) + O(\varepsilon^2),$$

so the relative error in κ is $\varepsilon/(S_b - \beta)$ to first order. Misestimation of the floor inflates powers of events acting near the floor and leaves powers of events acting far from it nearly unchanged.

Proof. Substituting β' into Theorem 4.1 leaves the numerator $S_b - S_a$ unchanged and replaces the denominator by $S_b - \beta - \varepsilon$. The stated ratio and its expansion follow. $\square$

Theorem 5.10 is the practical reason Theorem 3.8 matters. An analysis that sets the floor to the sample minimum systematically overestimates it—the sample minimum is an upper bound on the true infimum—and therefore systematically inflates the powers of exactly those events that operate closest to the floor, which are the events of greatest interest.

6 The telescoping obstruction

We now isolate the central methodological result of the paper. The composition law suggests an immediate test: decompose an observed process into events, compute each event’s power, form the multiplicative prediction, and compare it against the directly measured net power of the cascade. We show that under the most natural implementation this comparison is an identity.

Definition 6.1 (Instance-specific estimation). Powers are estimated *instance-specifically* when, for each cascade, each $\kappa(\gamma_i)$ is computed from Theorem 4.1 using the same observed uncertainties S_i, S_{i+1} that determine the cascade’s own measured net power.

Theorem 6.2 (Telescoping obstruction). *Under instance-specific estimation, the multiplicative prediction and the measured net power are equal for every cascade, identically:*

$$1 - \prod_{i=1}^n (1 - \kappa(\gamma_i)) = \frac{S_1 - S_{n+1}}{S_1 - \beta} = \kappa_{\text{measured}}.$$

Consequently any goodness-of-fit statistic comparing the two—a Pearson correlation, a root-mean-square error, a regression slope—takes its degenerate value ($r = 1$, RMSE = 0, slope 1) on every data set whatsoever, independently of the process generating the data.

Proof. This is Theorem 5.2 read as an identity rather than a prediction. With $g_i = S_i - \beta$, instance-specific estimation gives $1 - \kappa(\gamma_i) = g_{i+1}/g_i$ exactly, so the product telescopes:

$$\prod_{i=1}^n \frac{g_{i+1}}{g_i} = \frac{g_{n+1}}{g_1},$$

whence $1 - \prod_i(1 - \kappa_i) = 1 - g_{n+1}/g_1 = (S_1 - S_{n+1})/(S_1 - \beta)$, which is the measured net power. The two quantities are the same function of the data, so their difference vanishes pointwise and every discrepancy statistic attains its degenerate value. $\square$

Corollary 6.3 (The degenerate test has no null). *There is no data set, and no generating process, under which the instance-specific test fails. It therefore has no power against any alternative, and a reported agreement carries no evidential weight.*

Proof. By Theorem 6.2 the agreement holds identically, in particular for data generated by any process whatsoever, including processes violating every substantive claim of the algebra. A test that cannot fail cannot discriminate. $\square$

Remark 6.4 (Why this is easy to miss). The obstruction is invisible at the level of the code that implements the test: the two quantities are computed by different routines, one looping over constituent events and multiplying, the other differencing the endpoints of the cascade, and they agree to machine precision. The agreement looks like a striking confirmation of the composition law across thousands of cascades. It is instead a restatement of Theorem 5.2. We record this explicitly because an r of exactly 1.000 is precisely the outcome that should prompt suspicion rather than satisfaction, and because the same structure will arise in any framework whose composition law is a telescoping product.

6.1 The corrected test

The obstruction arises because the constituent powers and the cascade outcome are functions of the same numbers. Breaking the identity requires estimating powers from a source independent of the particular cascade being predicted. The natural such source is the *type* of an event.

Definition 6.5 (Event type; type-averaged power). Let each event carry a *type* $t(\gamma)$ drawn from a finite label set—for instance the ordered pair of state labels it connects. For a corpus of observed events, the *type-averaged power* of type θ is

$$\bar{\kappa}_\theta = \frac{1}{|\mathcal{T}_\theta|} \sum_{\gamma \in \mathcal{T}_\theta} \kappa(\gamma), \quad \mathcal{T}_\theta = \{\gamma : t(\gamma) = \theta\}.$$

The *type-averaged prediction* for a cascade $\gamma_1, \dots, \gamma_n$ is $\hat{\kappa} = 1 - \prod_{i=1}^n (1 - \bar{\kappa}_{t(\gamma_i)})$.

Theorem 6.6 (The type-averaged test is non-degenerate). *Suppose some type θ has at least two instances with distinct powers. Then there exist cascades for which $\hat{\kappa} \neq \kappa_{\text{measured}}$, and the map from data to the discrepancy $\hat{\kappa} - \kappa_{\text{measured}}$ is not identically zero. Hence the type-averaged test has a genuine null hypothesis and can fail.*

Proof. Let θ have instances with powers $\kappa \neq \kappa'$, so $\bar{\kappa}_\theta$ differs from at least one of them; without loss $\bar{\kappa}_\theta \neq \kappa$. Consider a cascade of length one consisting of that instance. Then $\kappa_{\text{measured}} = \kappa$ while $\hat{\kappa} = \bar{\kappa}_\theta \neq \kappa$, so the discrepancy is nonzero. Since the discrepancy is nonzero on this instance, it is not identically zero as a function of the data, and the equality $\hat{\kappa} = \kappa_{\text{measured}}$ is a substantive restriction rather than an identity. $\square$

Remark 6.7 (What the corrected test asks). The type-averaged test asks a genuine question: *is the catalytic power a property of the event type, stable across instances, such that composing type-level powers predicts instance-level outcomes?* A positive answer supports both the composition law and the claim that the chosen typing carves the process at its joints. A negative answer is informative in two distinguishable ways: the typing may be wrong—events of the same nominal type may not share a power—or the decomposition into events may be wrong. Section 7 shows how to tell these apart.

Proposition 6.8 (Exact bias of the type-averaged prediction). *Write $\kappa_i = \bar{\kappa}_{t(i)} + \delta_i$, where δ_i is the deviation of instance i from its type mean. Then*

$$\frac{1 - \kappa_{\text{measured}}}{1 - \hat{\kappa}} = \prod_{i=1}^n \left(1 - \frac{\delta_i}{1 - \bar{\kappa}_{t(i)}}\right),$$

so the prediction is exact if and only if the deviations vanish, and to first order in small δ ,

$$\hat{\kappa} - \kappa_{\text{measured}} \approx (1 - \hat{\kappa}) \sum_{i=1}^n \frac{\delta_i}{1 - \bar{\kappa}_{t(i)}}.$$

Proof. By Theorem 5.2, $1 - \kappa_{\text{measured}} = \prod_i (1 - \kappa_i) = \prod_i (1 - \bar{\kappa}_{t(i)} - \delta_i) = \prod_i (1 - \bar{\kappa}_{t(i)}) \prod_i (1 - \delta_i / (1 - \bar{\kappa}_{t(i)}))$, and $\prod_i (1 - \bar{\kappa}_{t(i)}) = 1 - \hat{\kappa}$ by Theorem 6.5. Dividing gives the stated ratio. Expanding the product to first order in δ yields $1 - \sum_i \delta_i / (1 - \bar{\kappa}_{t(i)})$, and substituting gives the approximation. $\square$

Theorem 6.8 makes precise what the corrected test measures: the discrepancy is driven by the within-type deviations δ_i , amplified by $1/(1 - \bar{\kappa})$ for high-power types. It also shows the deviations enter additively at leading order, so that cascades whose deviations cancel will be predicted well even when individual events are poorly typed—a source of optimism to be guarded against by examining cascade length.

7 When the corrected test has power

A non-degenerate test may still be uninformative if the data cannot distinguish its hypotheses. We characterise the regime in which the type-averaged test discriminates.

Definition 7.1 (Variance decomposition). For a corpus of events with types, let $\text{Var}_{\text{within}}$ denote the mean over types of the variance of κ within a type, and $\text{Var}_{\text{between}}$ the variance across types of the type means $\bar{\kappa}_\theta$. The *type separation ratio* is

$$\eta = \frac{\text{Var}_{\text{between}}}{\text{Var}_{\text{between}} + \text{Var}_{\text{within}}} \in [0, 1].$$

Proposition 7.2 (Degeneracy at vanishing separation). *If $\eta = 0$ —all type means coincide—then the type-averaged prediction is the same constant for every cascade of a given length, its correlation with the measured net power is undefined (zero variance in the predictor), and the test cannot distinguish the multiplicative law from any competitor that is also a function of the type means alone.*

Proof. If all $\bar{\kappa}_\theta$ are equal to a common $\bar{\kappa}$, then $\hat{\kappa} = 1 - (1 - \bar{\kappa})^n$ depends only on the cascade length n . Among cascades of fixed length the predictor is constant, so its sample variance is zero and the Pearson correlation is undefined. Any alternative law computed from the same type means is likewise constant at fixed length, hence indistinguishable. $\square$

Remark 7.3 (The observable that fails). Theorem 7.2 identifies the failure mode to check before interpreting any negative result. If the chosen observable maps every state label into a narrow band—so that all types have nearly equal mean power—then $\eta \approx 0$ and the test is underpowered by construction, regardless of whether the composition law holds. A negative result in this regime is evidence about the observable, not about the law. Section 10 exhibits exactly this situation in one of the three constructed regimes, and we report it as an absence of power rather than as a disconfirmation.

Proposition 7.4 (Competing composition laws). *The following are distinct functions of the constituent powers and coincide only in degenerate cases: the multiplicative law $1 - \prod_i (1 - \kappa_i)$;*

the additive law $\min(\sum_i \kappa_i, 1)$; the geometric-mean law $1 - (\prod_i (1 - \kappa_i))^{1/n}$; and the maximum law $\max_i \kappa_i$. For $n \geq 2$ and powers in $(0, 1)$ they satisfy

$$\max_i \kappa_i \leq 1 - \prod_i (1 - \kappa_i) \leq \min\left(\sum_i \kappa_i, 1\right),$$

with both inequalities strict unless some $\kappa_i = 0$; and the geometric-mean law is strictly below the multiplicative law for $n \geq 2$ whenever $\prod_i (1 - \kappa_i) < 1$.

Proof. For the left inequality, $\prod_i (1 - \kappa_i) \leq 1 - \max_i \kappa_i$ since every factor is at most 1 and one factor equals $1 - \max_i \kappa_i$; negating gives the claim, with equality iff all other factors are 1, i.e. all other powers vanish. For the right, expanding $\prod_i (1 - \kappa_i) \geq 1 - \sum_i \kappa_i$ (Weierstrass) gives $1 - \prod_i (1 - \kappa_i) \leq \sum_i \kappa_i$, with equality iff all cross terms vanish, i.e. at most one power is nonzero; combining with $\kappa_{\text{net}} \leq 1$ gives the stated bound. For the geometric mean, with $P = \prod_i (1 - \kappa_i) < 1$ we have $P^{1/n} > P$ for $n \geq 2$, so $1 - P^{1/n} < 1 - P$. $\square$

Theorem 7.4 supplies the comparison set for the empirical test: the four laws are ordered and distinct, so a corpus with adequate type separation can in principle select among them. It also shows that the multiplicative law is intermediate, which matters for interpretation—a finding that it outperforms both the additive and the maximum law is a two-sided result, not a one-sided one.

8 Closure

A cascade must terminate. The default criterion—stop when the attained uncertainty falls below a threshold—is inadequate for a reason we can now make precise.

Definition 8.1 (Event registry; reachable set). An *event registry* Γ is a finite set of events available to an observer. Fixing an initial state, let $\mathcal{C}(\Gamma')$ denote the set of terminal states reachable by cascades drawn from an invoked subset $\Gamma' \subseteq \Gamma$, partitioned into equivalence classes under a specified indistinguishability relation on terminal states.

Definition 8.2 (Closure). A cascade is *closed* at stage Γ' if for every event $\gamma \in \Gamma \setminus \Gamma'$ not yet invoked, adjoining γ adds no new equivalence class: every terminal state reachable using γ lies in a class already present in $\mathcal{C}(\Gamma')$.

Theorem 8.3 (Closure is strictly stronger than a threshold). *For every threshold θ on attained uncertainty there exist registries and initial states such that a cascade attains uncertainty below θ while failing to be closed: an uninvoked event leads to a terminal state in a distinct equivalence class. Conversely, closure does not entail any particular attained uncertainty. The two criteria are therefore not ordered by implication in the direction threshold $\Rightarrow$ closure, and closure is the stronger requirement in the sense that it can fail when the threshold is met.*

Proof. Construct a registry with two events γ_A, γ_B leading from the initial state to terminal states x_A, x_B respectively, with x_A and x_B in distinct equivalence classes, and with the uncertainty attained at x_A below θ . Invoking γ_A alone attains uncertainty below θ , satisfying the threshold criterion. But γ_B is uninvoked and reaches x_B , whose class is absent from $\mathcal{C}(\{\gamma_A\}) = \{[x_A]\}$; hence the cascade is not closed by Theorem 8.2. For the converse direction, a registry in which every event leads to the same class is closed after one invocation regardless of the uncertainty attained, which may exceed θ . $\square$

Remark 8.4 (The failure mode closure prevents). The construction in Theorem 8.3 is not pathological; it is the ordinary situation of a single line of evidence that is internally consistent. Any one coherent line of inquiry drives its own uncertainty down and satisfies any fixed threshold, precisely because it is internally consistent. What a threshold cannot detect is that a second, independent line would have terminated elsewhere. Closure detects exactly this, because it quantifies over the events *not* invoked. The criterion is not that the current answer looks good but that the space of answers has stopped producing new destinations.

Theorem 8.5 (Termination). *Over a finite registry Γ , closure is reached after finitely many invocations.*

Proof. Each invocation either adds a new equivalence class to $\mathcal{C}$ or does not. The number of equivalence classes is bounded by the number of terminal states reachable from a finite registry, which is finite. Hence only finitely many invocations can add a class. After the last such invocation, every remaining event adds no class, which is Theorem 8.2. In the worst case every event of Γ is invoked, giving termination in at most $|\Gamma|$ steps. $\square$

Theorem 8.6 (Dichotomy of outcomes). *At closure, exactly one of the following holds:*

- (i) **Convergent closure:** $\mathcal{C}$ contains exactly one equivalence class, and the cascade reports its representative;
- (ii) **Contested closure:** $\mathcal{C}$ contains two or more classes, and no single terminal state is supported by the registry; the honest report is the plurality of classes together with the events leading to each.

Proof. By Theorem 8.5 closure is reached, and $\mathcal{C}$ is then a nonempty finite partition of the reachable terminal states—nonempty because at least one cascade is admissible by hypothesis. A finite nonempty partition has either exactly one block or at least two, and these are mutually exclusive and exhaustive. In case (i) every admissible cascade terminates in the same class and the representative is well defined up to indistinguishability. In case (ii) two admissible cascades terminate in distinct classes; since both are admissible, the registry supports neither over the other, and selecting one would assert a discrimination the data do not license. $\square$

Corollary 8.7 (Declining is a defined outcome). *An observer operating under closure has a well-defined behaviour when its evidence is genuinely conflicting: report contested closure. This is not a failure of the procedure but one of its two normal terminations.*

Remark 8.8 (Relation to stopping rules based on information gain). Sequential design procedures typically stop when the expected gain from a further observation falls below a cost [3, 7]. Such a rule is a threshold rule in the sense of Theorem 8.3: it evaluates a magnitude attached to the current state and compares it against a constant. Closure evaluates instead whether the *set* of reachable outcomes is stable under the remaining available events, which is not a magnitude comparison and is not obtained by lowering a threshold. The two can disagree in both directions: a cascade may have negligible expected gain yet be unclosed, and may be closed while substantial gain remains available within an already-reached class.

9 Three readings of a residual

The residual admits three distinct readings depending on how its link to the next event is treated. We record them because they determine what an instrument should log.

Definition 9.1 (Committed record). A cascade’s *committed record* is the count of events it has performed. An event is *committed* when its post-state has been entered.

Proposition 9.2 (Residual as propagator). *An event γ_{i+1} acts on a state distinct from that on which γ_i acted if and only if γ_i left a nonzero residual change, i.e. $\kappa(\gamma_i) \neq 0$. A cascade of events with vanishing powers revisits the same state and commits no distinct events.*

Proof. The state after γ_i differs from the state before exactly when $S_{i+1} \neq S_i$, which by Theorem 4.1 holds iff $\kappa(\gamma_i) \neq 0$. If $\kappa(\gamma_i) = 0$ the pre- and post-states have equal uncertainty and γ_{i+1} acts on a state indistinguishable, in the algebra, from its predecessor. $\square$

Proposition 9.3 (Monotone record). *The committed record is non-decreasing along a cascade and strictly increases at each committed event. Two states with equal uncertainty but different records are distinguishable by the record; recurrence of a value of S is therefore not a return to a previous state of the cascade.*

Proof. Each committed event increments the count by one by Theorem 9.1. To decrease the count an event would have to un-commit a prior event, which is itself an event and increments the count. Hence monotone. Two states with equal S and records $r < r'$ are separated by the predicate “record $\leq r$ ”, hence distinct. $\square$

Proposition 9.4 (Three readings). *A residual admits exactly three readings according to the treatment of its link to the successor event:*

- (i) *as time, when read as the increment of the committed record (Theorem 9.3);*
- (ii) *as information, when the link from a residual to the specific successor it enabled is recorded, so that the cascade is specified as a determinate map;*
- (iii) *as entropy, when residuals are present but their links are not recorded, so that the cascade is known only through a distribution over which residual enabled which successor, of Shannon content $H = -\sum_j p_j \log p_j$ [10].*

Recording links converts (iii) into (ii); discarding them reverses the conversion.

Proof. (i) is Theorem 9.3. (ii) A recorded link is a determinate assignment from each residual to its successor; specifying all such assignments specifies the cascade exactly. (iii) With links unrecorded, the cascade is described by the distribution over possible assignments, whose uncertainty is by definition the Shannon entropy of that distribution [4, 10]. Recording an assignment removes it from the distribution and makes it determinate, converting entropy into information; discarding a record has the inverse effect. $\square$

Remark 9.5 (Consequence for instrumentation). Theorem 9.4 states precisely what provenance logging buys. An apparatus that records which prior result enabled which subsequent step converts an entropic description into an informational one; an apparatus that records only the sequence of states retains the temporal reading and loses the other two. The choice is not stylistic: the composition law can be tested only against cascades whose event sequence is recorded, and the type-averaged correction of Section 6 additionally requires that each event’s type be recorded.

10 Numerical verification

Every claim above is proved, so numerical work serves a different purpose: to confirm that the implementation of the algebra behaves as the theorems require, to exhibit the telescoping obstruction concretely, and to demonstrate the regime in which the corrected test loses power. We report all three, including the negative result.

10.1 Method

We construct synthetic cascades with known generating structure. Each instance fixes a floor $\beta > 0$, an initial uncertainty S_1 , and a set of event types with assigned mean powers and within-type dispersion. Cascades are generated by drawing events of specified types, applying each event’s realised power to the current gap, and recording the resulting state sequence. Three regimes are used, differing only in type separation (Theorem 7.1): a *separated* regime with well-spaced type means, an *intermediate* regime, and a *compressed* regime in which all type means are nearly equal. The compressed regime instantiates Theorem 7.3.

For each regime we compute: the identity of Theorem 6.2 under instance-specific estimation; the agreement between type-averaged predictions and measured outcomes under each of the four laws of Theorem 7.4; the type separation ratio η ; and the behaviour of the floor estimator of Theorem 3.8 as the sample grows. Full numerical results are recorded in machine-readable form alongside this paper.

10.2 What the computations establish

The verification is designed to be capable of failure at four points, and we state the expected outcome at each so that a discrepancy would be visible. First, the telescoping identity of Theorem 6.2 must hold to machine precision on every generated cascade, in every regime, including regimes generated by processes that violate the type-stability hypothesis: the identity is algebraic and cannot be sensitive to the generating process. Second, the type-averaged test of Theorem 6.5 must *not* reproduce that identity: its discrepancy must be nonzero, as Theorem 6.6 requires. Third, in the separated regime the multiplicative law must outperform its three competitors, since the data are generated multiplicatively; a failure here would indicate an error in the estimator rather than in the algebra. Fourth, in the compressed regime the test must lose discriminating power, with $\eta \approx 0$ and the four laws becoming indistinguishable—this is the predicted absence of power of Theorem 7.2, and reporting it as a disconfirmation of the composition law would be an error of exactly the kind Theorem 7.3 warns against.

Additionally, the floor estimator is checked against the two alternatives of Theorem 3.8: the sample-minimum estimator, which is positive by construction and therefore incapable of falsifying a positivity claim, and the asymptotic estimator, which can return a value indistinguishable from zero when the generating process has no floor. We verify three things: that the sample minimum is strictly positive on both a floored and an unfloored process, so that its positivity is uninformative; that it is biased upward on the floored process, with the consequence for near-floor powers given by Theorem 5.10; and that the asymptotic estimator recovers the true floor on the floored process while returning approximately zero on the unfloored one. We note explicitly that with very large samples the sample minimum may nonetheless *order* the two processes correctly; ordering is not falsification, and the claim under test is the latter.

11 Scope and limitations

11.1 What is proved

The results of Sections 3 to 5 are theorems about the definitions: given that catalytic power is a fraction of an outstanding gap measured from a fixed floor, composition is multiplicative, the floor is unattainable in finitely many steps, and convergence is governed by summability. The results of Sections 6 and 7 are theorems about estimation: instance-specific testing is degenerate, type-averaged testing is not, and the latter’s power is governed by type separation. The results of Section 8 are theorems about termination.

11.2 What is assumed

Axiom 1 is the sole substantive hypothesis, and it is used only to derive the positivity of the floor. A reader who prefers to posit a positive floor directly may do so and retain every subsequent result; what is lost is the account of *why* the floor is positive and the falsifiable prediction of Theorem 3.8.

11.3 What is not claimed

We do not claim that any particular physical, biological, or computational process decomposes into events whose powers are type-stable. That is an empirical question, distinct for each substrate,

and the apparatus of Section 6 exists precisely to make it answerable rather than to presuppose an answer. We do not claim that the multiplicative law is the correct description of any given process; we claim that it is the law implied by the definitions, that a specific naive test of it is vacuous, and that a corrected test exists whose power is characterised. Finally, the composition law governs cascades over a *fixed* floor; processes in which the floor itself moves—because the observer’s situation changes qualitatively—are outside the scope of Theorem 5.2 as stated, and require the floor to be re-derived for each regime.

11.4 Conclusion

The algebra has one quantity and one law. The quantity is the residual, whose positivity is forced by the impossibility of completing an identification against a whole that no finite stage exhausts. The law is that residuals multiply, from which the composition rule, the unattainability of the floor, and the summability criterion for convergence all follow. The methodological content of the paper is negative and, we think, more useful than the positive content: the natural test of the composition law is an identity in disguise, it will report perfect agreement on any data whatsoever, and only a test that estimates powers independently of the outcome it predicts has a null hypothesis at all. The termination criterion completes the apparatus by replacing a threshold, which a single self-consistent line of evidence can always satisfy, with a condition on the stability of the reachable set—and by making a declaration of contested evidence one of the two normal ways an inquiry ends.

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